

CE 506: Analysis of Indeterminate Structures

Course Code: CE 506

Course Name: Analysis of Indeterminate Structures

L-T-P-C: 3-0-0-3

Prerequisite: CE 301 - Strength of Materials and Structures

Students intended for: UG

Elective or Core: Elective

Approval: 9th Senate

Course contents

- **Introduction**

Statically determinate vs. statically indeterminate structures, Determining degrees of static and kinematic indeterminacy, Methods of analysis. [3 Lectures]

- **Moment distribution method**

Analysis of indeterminate beams subjected to loads and uneven settlement of supports, analysis of rigid frames with and without side sway. [6 Lectures]

- **Three-moment theorem**

Derivation and application for analysis of continuous beams subjected to loads and uneven settlement of supports. [6 Lectures]

- **Module IV**

Kani's method and column analogy method for frames with and without sway. [6 Lectures]

- **Analysis using Force methods**

Analysis of indeterminate beams and frames, law of reciprocal deflections, theorem of least work, Analysis of indeterminate trusses considering reactions and/or member forces as redundant, reactions due to yielding of supports. [6 Lectures]

- **Approximate analysis of statically indeterminate structures**

Analysis of trusses, vertically loaded building frames, Portal and Cantilever methods for laterally loaded building frames. [6 Lectures]

- **Introduction to matrix methods for structural analysis**

Flexibility and stiffness matrices, analysis of continuous beams, rigid and pin jointed frames. [9 Lectures]

Text Books

1. C.S.Reddy, **Basic Structural Analysis**, Tata McGraw Hill, New Delhi, 2001.
2. C.K. Wang, **Intermediate Structural Analysis**, Tata McGraw Hill, New Delhi, 2010.
3. DR.C. Hibbeler, **Structural Analysis**, 6th Edition, Pearson Education, 2009.
4. Ghali, A., Neville, A.M. and Brown, T.G., **Structural analysis: a unified classical and matrix approach**, Spon press, USA and Canada, 2003.

5. C.H. Norris, J.B. Wilbur, S.Utku, **Elementary Structural Analysis**, Tata McGraw Hill, 2003.
6. L. S. Negi and R. S. Jangjid, **Structural Analysis**, Tata Mc. Graw, New Delhi, 1997.
7. Pandit G S and Gupta S P, **Structural Analysis: a matrix approach**, Tata McGraw Hill, New Delhi, 2001.
8. Natrajan, C. and Revathi, P, **Matrix methods of Structural Analysis: Theory and Problems**, PHI, New Delhi, 2014.

Reference Books:

1. Mau, S.T., **Introduction to structural analysis: Displacement and force methods**, CRC press, FL, 2012.
2. AWeaver, W. Jr. and Gere, J.M., **Matrix analysis of framed structures**, Springer, 2012.
3. Ranzi, G. and Raymond, I.B., **Structural analysis: principles, methods and modelling**, CRC press, FL, 2014.